

p-Chalk

Film breakdown that knows your playbook

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Project Overview

High school football coaches spend hours manually tagging game and practice film before they can begin self-scouting and game planning. Existing platforms organize film, but they generally do not understand each program's playbook or automatically translate what appears on screen into the coach's own terminology. Chalk is a web application that connects those two sources: coaches enter or draw their plays, upload film, and receive a play-by-play breakdown mapped to their formations, motions, and route concepts. Low-confidence predictions are flagged for review, and coach corrections become structured feedback for future improvement.

The project already has a working full-stack prototype and an experimental computer-vision pipeline. The P-ai team will turn those pieces into a measured end-to-end product. We will use field-keypoint detection and homography estimation to project broadcast or sideline footage onto a standardized 2D field; object detection, multi-object tracking, and appearance clustering to follow players and separate teams; temporal and geometric features to detect snaps, formations, motion, and routes; and similarity-based retrieval or classification to match extracted play representations to a team's playbook. Our goal is a coach-facing pilot that processes offensive self-scout film, produces interpretable route cards and play labels, reports calibrated confidence, supports human correction, and exports usable breakdown data.

Tech Stack

- Languages: Python, TypeScript, and SQL
- Backend and frontend: FastAPI, SQLAlchemy, Next.js, and React
- Computer vision and ML: OpenCV, NumPy, PyTorch, Hugging Face/SigLIP, RF-DETR or another permissively licensed detector, ByteTrack, and a permissively licensed re-identification model

- Methods: homography estimation, geometric computer vision, clustering, nearest-neighbor retrieval, supervised classification, and confidence calibration
- Data and infrastructure: SQLite locally; Postgres/Supabase and Cloudflare R2 or equivalent object storage for production
- Quality and collaboration: Pytest, Vitest, Playwright, GitHub, GitHub Issues/Projects, code review, and continuous integration

Dataset

The primary dataset will be game and practice film voluntarily provided by partner football programs, paired with each program's playbook and coach-approved labels for snap time, possession, formation, player roles, motion, and routes. Students will use Chalk's existing annotation interfaces and written labeling protocol to create train, development, and sealed holdout splits; corrections made during review will be stored as additional team-specific labels. Because the film may depict minors and reveal competitive strategy, raw video will remain access-controlled, will not be committed to GitHub, and will be retained only as long as needed. Initial models will rely on permissively licensed pretrained vision models and small, program-specific adaptation or retrieval rather than requiring a large public football dataset. Evaluation will report route recall and precision, registration quality, confidence calibration, and end-to-end coach correction time, with results separated by lighting, camera angle, field quality, and program.

Ethical Considerations

The initial dataset will be small and concentrated in the camera setups, uniforms, field conditions, and offensive systems of participating programs, so performance may not generalize across schools; we will use program-level holdouts, subgroup error analysis, and explicit low-confidence abstention rather than presenting uncertain outputs as facts. Because the footage may include minors and confidential strategy, participation must be authorized by the program, access will be least-privilege, identifiable video will not be placed in public repositories, and retention and deletion procedures will be documented. Chalk could reduce repetitive analyst labor and expand access for under-resourced teams, but it could also enable unauthorized opponent scouting or surveillance, so the pilot is limited to consented, offense-only self-scouting with human review before export.

Team Structure

- Target Team Size: 4-6 students (not including PMs)
- Subteam Structure: A six-person project team led by PM Aidan Chen and team leads Kavin Prasanna and Connor Wang, organized into technical and nontechnical workstreams with shared ownership of evaluation and product direction
 - Technical contributors will work on CV/ML, data, backend, and frontend tasks. Design, research, and growth contributors can own visual identity,

coach interviews, school outreach, competitor analysis, documentation, and workflow testing. Mixed-discipline pairs will integrate these outputs into the product each week.

- **Organizational & Collaboration Frameworks:** A weekly meeting for high-level goals and shared challenges; brief asynchronous standups on Mondays and Wednesdays; and an in-person Friday working session when schedules allow. GitHub Projects and Issues will track scoped work, while Discord or Slack will keep Aidan, Kevin, and Connor easy to reach for questions, ideas, and blockers.
 - The three leads will maintain clear goals and acceptance criteria while giving members creative ownership over how they solve problems. Weekly demos, peer feedback, documented decisions, tests for technical work, and visible recognition of design, research, and outreach contributions will keep the team coordinated and inclusive.

Timeline

The ten-week plan prioritizes a demonstrable vertical slice before model optimization. Both subteams integrate at the end of every two-week phase, and metrics plus coach-facing artifacts determine whether work advances.

Phase	Deliverables and ownership
Weeks 1–2	Baseline and product contract — CV/ML reproduces current registration/route metrics and defines acceptance tests; Product/Platform maps the coach workflow and integrates one sample play end to end.
Weeks 3–4	Data and evaluation pipeline — CV/ML improves annotation tools and establishes train/development/hold out splits; Product/Platform builds secure ingestion and

	metadata flows.
Weeks 5–6	Route extraction vertical slice — CV/ML connects registration, tracking, team classification, and route traces; Product/Platform integrates jobs, progress, and visual route cards.
Weeks 7–8	Playbook mapping and review loop — CV/ML implements similarity matching and calibrated confidence; Product/Platform completes correction workflows and Hudl-format CSV export.
Weeks 9–10	Pilot readiness and final demo — Both subteams test on held-out film, document limitations and privacy procedures, package deployment and onboarding, and present metrics plus a coach-facing workflow.

Project Member Requirements

The only requirement is interest in either sports, CV, data, or entrepreneurship! We are a team that has ambitious visions, but also wants to have fun and learn a lot along the way. Everyone is welcome and encouraged to apply. Programming experience is not required for every role. Technical contributors should be comfortable with Python or TypeScript and an introductory computer science course; experience with computer vision, machine learning, React, FastAPI, or statistics is helpful but not required. Nontechnical contributors can bring visual design, digital media, communications, market research, interviewing, outreach, social media, writing, or football knowledge.

Every member must handle sensitive film responsibly, document their work, seek and give feedback, and communicate blockers early. Estimated time commitment of 4-6 hours per week.